Test Assignment – Senior WPF/Avalonia Developer

Program Name

PCB Manufacturing

Purpose and Scope

This program simulates creation of a printed circuit board (PCB) manufacturing order. It is a **test assignment** for candidates applying for **WPF** or **Avalonia** developer roles. All data may be predefined and stored locally. The goal is to evaluate: - Practical knowledge of UI development - MVVM architecture - Ability to build clean, maintainable code - Ability to work with layout systems, styles, validation, templating, and graphics - Basic understanding of modular application structure - Understanding of cross-platform development (for Avalonia track)

The application must consist of **one screen** with **three tabs**: **Preferences**, **Quote**, **Order**.

The candidate may choose: - **WPF (.NET 6+ or .NET Framework)**, or - **Avalonia (.NET 8 recommended**.

Source code must be structured using **MVVM**, feature-based project organization is encouraged.

General Technical Requirements

Architecture

- Strict MVVM (no code-behind except view-only helpers)
- Use of `INotifyPropertyChanged` (or source generators)
- Use of Commands (RelayCommand, DelegateCommand, ReactiveCommand, etc.)
- Clear separation of Views, ViewModels, Models, Services
- For Avalonia: use DataTemplates for VM-to-View mapping

UI / UX Requirements

- Responsive layout
- Use of Styles and Control Templates where appropriate
- Consistent color theme
- Element spacing, alignment, and accessibility should be reasonable

Validation

- Postcode validation must use:
    - DataAnnotations **or** custom validation rules (WPF)

    - `IDataErrorInfo` / `INotifyDataErrorInfo`
    - For Avalonia: `IDataValidationPlugin`, DataAnnotations, or custom logic
- Invalid input must show visual feedback

Data

- Predefined materials, colors, and parameters stored in:
    - JSON file **or** static in-code structures

Bonus Points

- Unit tests (NUnit/xUnit)
- DI container usage (Microsoft.Extensions.DependencyInjection)
- MVVM frameworks (Prism, ReactiveUI, CommunityToolkit.MVVM)
- Animations or transitions
- Cross-platform themed UI (Avalonia)

Application Requirements

1. **Preferences Tab**

Required features:

1. **Dropdown lists** with 2–3 predefined values each:
    - Material selection
    - Solder mask color selection
    - Board thickness (optional bonus)
2. **Material selection** must influence:
    - Price modifier
    - Production time
3. **Postcode field**:
    - Must be validated
    - Must show visual indicator of invalid state
4. **Dynamic summary panel** showing selected preferences (optional bonus)

Technical expectations

- Use of `ComboBox`, `Binding`, `SelectedItem`
- Use of converters if needed
- Validation through MVVM

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2. **Quote Tab**

Required features:

1. **Parameter Table**:
    - Table shows grouped predefined values
    - Example groups: Dimensions, Layers, Finish Type
    - Should be displayed in a `DataGrid`, `ItemsControl`, or custom template
2. **Parameter Preview Panel**:
    - Graphical representation of PCB parameters
    - Example: simple rectangle with color applied based on selected solder mask
    - Should update when Preferences change

Technical expectations

- Use of `ObservableCollection` or static collections
- Use of bindings to update graphical panel
- For Avalonia: use `Canvas`, `Border`, or `Path`

Bonus

- Implement vector drawing using `Geometry`
- Implement dynamic layout

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3. **Order Tab**

Required features:

1. **Order Button**:
    - Must validate all input data
    - Must show summary dialog or navigate to confirmation state
    - Logic implemented in ViewModel

Technical expectations

- Use of `ICommand`
- Use of async handling for button click (bonus)
- Use of dialog service abstraction (bonus)

Project Structure Example

```
/PCBManufacturing
  /Models
  /ViewModels
```

```
/Views
/Services
/Resources (styles, templates)
App.xaml / App.axaml
```

Deliverables

Candidate must provide: 1. Full source code (GitHub or zip) 2. Short README including: - Build instructions - Framework versions - Architecture overview - If Avalonia: supported platforms tested 3. Screenshots of working UI 4. (Optional) Unit tests

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Evaluation Criteria

- Clean architecture (MVVM, low coupling)
- Code readability and organization
- XAML quality (styles, templates, layout correctness)
- Validation correctness and user experience
- Clean bindings and command implementation
- Quality of graphical representation
- For Avalonia: cross-platform correctness and control usage

Additional Optional Features (Extra Credit)

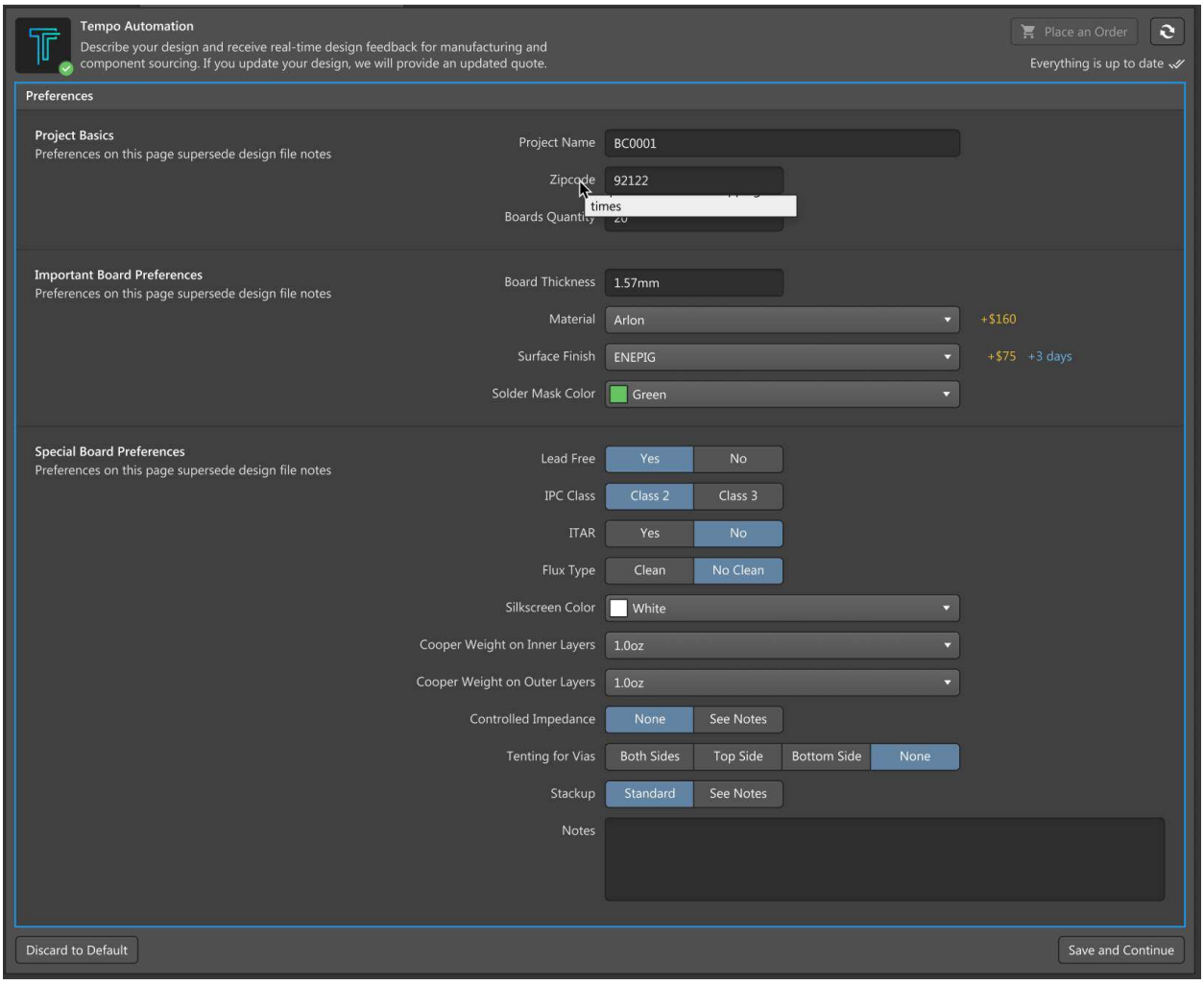
- Localization (RU/EN)
- Theme switching (light/dark)
- Saving preferences to JSON
- More advanced visuals (layers preview)
- Using vector icons instead of raster
- Applying ReactiveUI for Avalonia version
- Using CommunityToolkit.MVVM

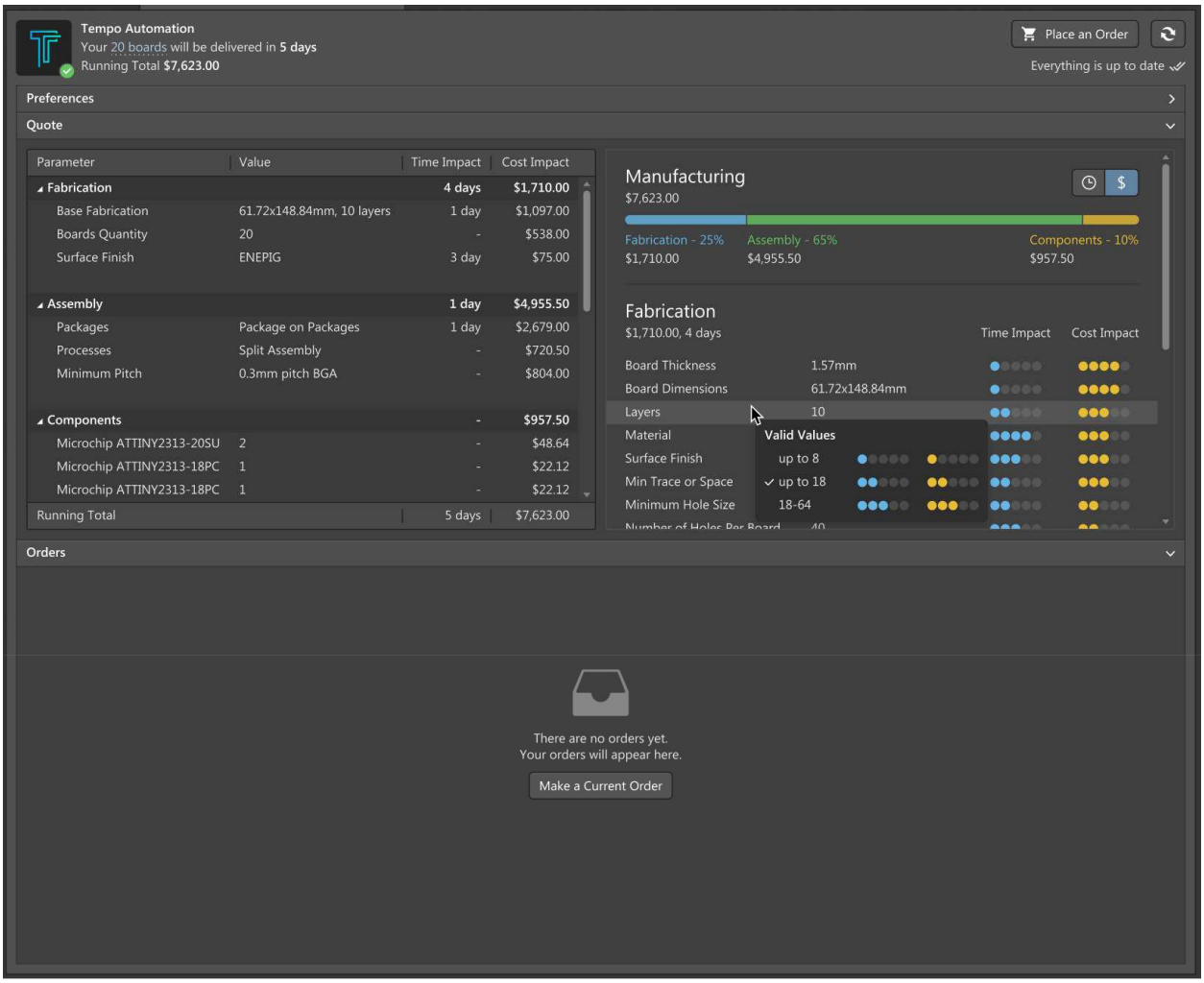
Notes

- The color scheme in the original PDF is not required to match.
- The candidate has full creative freedom in layout and visual style.

- All predefined data may be arbitrary.

Mockups




Tempo Automation
Your 20 boards will be delivered in 5 days
Running Total $7,623.00
Place an Order
Everything is up to date
Preferences
Quote
Parameter
Value
Time Impact
Cost Impact
Fabrication
4 days
$1,710.00
Base Fabrication
61.72x148.84mm, 10 layers
1 day
$1,097.00
Boards Quantity
20
-
$538.00
Surface Finish
ENEPIG
3 day
$75.00
Assembly
1 day
$4,955.50
Packages
Package on Packages
1 day
$2,679.00
Processes
Split Assembly
-
$720.50
Minimum Pitch
0.3mm pitch BGA
-
$804.00
Components
-
$957.50
Microchip ATTINY2313-20SU
2
-
$48.64
Microchip ATTINY2313-18PC
1
-
$22.12
Microchip ATTINY2313-18PC
1
-
$22.12
Running Total
5 days
$7,623.00
Manufacturing
$7,623.00
Fabrication - 25%
$1,710.00
Assembly - 65%
$4,955.50
Components - 10%
$957.50
Fabrication
$1,710.00, 4 days
Time Impact
Cost Impact
Board Thickness
1.57mm
Board Dimensions
61.72x148.84mm
Layers
10
Material
Surface Finish
Min Trace or Space
Minimum Hole Size
Valid Values
up to 8
up to 18
18-64
Orders
There are no orders yet.
Your orders will appear here.
Make a Current Order